Predicting Benchmark Scores from Unpaired Model Evaluations

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Abstract

Evaluating a generative model on all queries from all available benchmark tasks is prohibitively expensive. As such, models are often evaluated on different subsets of tasks and/or different subsets of queries from a given task. The resulting collection of responses and the corresponding (model, task) score matrix are unpaired and sparse. In this paper, we introduce the Product Kernel Perspective Space (PKPS), a collection of low-dimensional representations of models based on their embedded responses. PKPS compares responses to similar, rather than identical, queries, so models that answered different queries remain comparable; paired methods are recovered as a special case. Theoretically, we show that benchmark prediction using PKPS is query-efficient relative to methods that only use information from queries all models have responded to. Empirically, we show that PKPS outperforms relevant baselines on two tasks—query-efficient evaluation and (model, task) score matrix imputation—across two large benchmark suites.

1 Introduction

Generative models are evaluated on an ever-growing collection of benchmarks spanning knowledge, reasoning, code, translation, and safety [Liang et al., 2023]. Evaluating a model on all of them is often prohibitively expensive. As a result, models are evaluated on different subsets of tasks and on different subsets of queries within a task: a recent compilation of 84 frontier models on 133 benchmarks finds only 23% of the score matrix filled [Zeng and Papailiopoulos, 2026]. Within a curated benchmark, every model is forced to answer an identical set of queries—but this controlled setting is the exception. Arena-style platforms score each model on the conversations its users generate [Chiang et al., 2024]; contamination-aware benchmarks refresh their items, so models evaluated in different months answer different questions [White et al., 2025]; and cost-driven subsampling gives each evaluation run its own item subset [Perlitz et al., 2024]. The resulting (model $\times$ task) score matrix is sparse, its entries are noisy averages over few queries, and the responses that produced it are *unpaired*: comparing models in general requires using responses to different queries. Predicting the matrix's missing and true entries would let practitioners reuse cached evaluations instead of paying to re-run them [Polo et al., 2024, Vivek et al., 2024, Zeng and Papailiopoulos, 2026].

Two lines of work aim to reduce this cost by predicting scores rather than computing them. Efficient benchmarking extrapolates a task's full score from a small scored subset of its items, classically by item-response theory [Polo et al., 2024]; matrix completion imputes cells that were never run, from low-rank structure across the matrix [Zeng and Papailiopoulos, 2026]. These are two ends of the same problem: every cell of the score matrix is observed with some number of queries, possibly zero, and predicting its true score amounts to denoising when that number is small and to completion when it is zero. A single representation of each model that is accurate enough to predict any cell would serve both ends.

Neither line of work has such a representation: both use only each cell's aggregate score, and the responses that produced those scores carry more information than the scores themselves. The Data Kernel Perspective Space (DKPS) [Helm et al., 2026] uses this information: it embeds each model from its responses via classical multidimensional scaling and regresses benchmark scores in the resulting space. The DKPS distance between two models, however, is defined only over responses to shared queries—a *paired* design; in practice this restricts analysis to maximal subsets of models that share a common query set [Helm et al., 2026]. To our knowledge, no existing method predicts the score matrix from evaluation data as it is typically available: sparse, noisy, and unpaired.

Table 1: MAE of benchmark prediction methods on both suites and tasks (best per column bolded, second best underlined; setup described in Section 4.1): query-efficiency at budgets $m \in \{1,8\}$ queries per cell at full task coverage, and completion of missing cells at cohort sizes $n \in \{10, \max\}$ with task coverage p_{task} and query depth p_{query} fixed at 0.5, except for the sp. columns where p_{task}=0.2 and n=10. "—" denotes a setting where a method is not applicable.

	HELM suite (18 tasks, 93 models)					**EEE suite** (16 tasks, 45 models)				
	Query-eff.		Completion			Query-eff.		Completion		
Method	(m=1)	(m=8)	(n=10)	(n=93)	(sp.)	(m=1)	(m=8)	(n=10)	(n=45)	(sp.)
Sample score	0.292	0.092	—	—	—	0.224	0.081	—	—	—
IRT	0.348	0.343	—	—	—	0.165	0.139	—	—	—
LRMC	—	—	0.139	0.105	0.208	—	—	0.109	0.096	0.144
DKPS	0.168	0.109	—	—	—	0.116	0.077	—	—	—
PKPS	0.136	0.108	**0.120**	0.103	**0.171**	0.076	0.058	**0.082**	0.070	**0.108**
Ensemble	**0.125**	**0.073**	**0.120**	**0.099**	**0.171**	**0.073**	**0.049**	**0.082**	**0.069**	**0.108**

In this paper, we introduce the **Product Kernel Perspective Space (PKPS)**, a collection of low-dimensional representations of models based on their embedded responses. PKPS compares every response of one model to every response of another, weighting each comparison by the similarity of the queries that produced it: the affinity between two models is a product of a query kernel and a response kernel, normalized by the total query-kernel mass, so that two models are not deemed dissimilar merely because they answered different sets of queries (Figure 1). Models are embedded from these affinities by classical multidimensional scaling, and paired DKPS is recovered exactly when the query kernel is the identity (Section 3). Table 1 previews our main results.

Contributions. We make two primary contributions.

1. *Methodological & theoretical:* we extend the DKPS and its query-efficiency theory to the unpaired regime. Benchmark prediction in the PKPS is query-efficient relative to the sample score, with a query budget that does not depend on the fraction of queries shared across models, and relative to paired DKPS at every pairing fraction (Theorem 1 and Corollary 1).
2. *Empirical:* on two benchmark suites—93 models on 18 HELM tasks and 45 modern models on 16 tasks from the Every Eval Ever datastore [EvalEval Coalition, 2026]—we demonstrate utility on two problems: query-efficient evaluation, where PKPS matches the accuracy of the sample score with up to an 8× smaller query budget, and score-matrix completion, where ensembling PKPS with low-rank matrix completion reduces error by up to 28% (Section 4).

1.1 Problem setting and notation

A *model* is a random mapping $f : \mathcal{Q} \to \mathcal{X}$ from a query space to a response space; querying f at q returns a draw from the distribution $f(q)$. We observe n models $f_1, \dots, f_n$, each run on its own *query set* $Q_i \subseteq \mathcal{Q}$; the sets may overlap arbitrarily, and the paired design $Q_1 = \cdots = Q_n$ is a special case. Two fixed embedding functions map raw objects to vectors: a *response* embedding $\text{embed}_R : \mathcal{X} \to \mathbb{R}^p$ and a *query* embedding $\text{embed}_Q : \mathcal{Q} \to \mathbb{R}^{p'}$. For $q_j \in Q_i$ we write $x_{ij} = \frac{1}{r}\sum_{k=1}^{r} \text{embed}_R\big(f_i(q_j)^{(k)}\big)$ for the average of the r sampled response embeddings and $u_j = \text{embed}_Q(q_j)$ for the embedded query.

Queries are grouped into T tasks. Model i's *true score* on task t, y_{it}, is its mean score over the task's full query set; running the model on $m_{it} \geq 0$ of those queries yields an observed *sample score* whose noise scales as $1/\sqrt{m_{it}}$. The goal is to predict y_{it} for every cell of the (model × task) matrix: cells observed with few queries ($m_{it} > 0$; *denoising*, the efficient-benchmarking end) and cells never run ($m_{it} = 0$; *completion*). We write y_i when the benchmark has a single score. Two properties of the design recur throughout: *coverage*, the fraction of cells with $m_{it} > 0$, and *pairing*, the fraction ρ of a cell's queries shared across models; curated leaderboards sit near $\rho \approx 1$, while deployed evaluation sits near $\rho \approx 0$.

2 Related work

Low-dimensional representations of models. Some approaches construct representations by comparing internal activations across models [Duderstadt et al., 2023, Huh et al., 2024, Horwitz et al., 2025] or by comparing model weights directly [Chen et al., 2025]. Evaluation frameworks, however, typically permit only black-box access: they observe responses, not internals. The Data Kernel Perspective Space (DKPS), first introduced in the context of monitoring multi-agent systems [Helm et al., 2024], addresses this setting by mapping a collection of models into a low-dimensional Euclidean space via classical multidimensional scaling of average embedded responses [Torgerson, 1952]. Prior work established consistency and concentration of the representations [Acharyya et al., 2024, 2025], supervised inference thereon [Helm et al., 2025], and query efficiency [Helm et al., 2026]. These constructions compare responses to shared queries, a paired design. PKPS replaces the identity matching with a product kernel so that models evaluated on different queries remain comparable.

Benchmark prediction. A second line of work predicts unobserved (model, task) scores so that not every evaluation must be run; efficient benchmarking and matrix completion are its two ends. Polo et al. [2024] use item-response theory to construct small item subsets whose scores predict full performance [see also Rasch, 1960, Kipnis et al., 2025]; Vivek et al. [2024] cluster items by cross-model agreement and score only the cluster centers; Bean et al. [2025] select items using cognitive-scale embeddings; Li et al. [2024] select items adaptively via reinforcement learning; and Perlitz et al. [2024] analyze the cost–reliability tradeoff of benchmark designs. Relatedly, Ilyas et al. [2022] predict a model's outputs as a function of its training data, Prabhu et al. [2024] reuse past evaluations in lifelong benchmarks, and Helm et al. [2026] predict held-out scores from cached responses. On the completion end, Zeng and Papailiopoulos [2026] observe that the (model $\times$ task) score matrix is approximately rank-two and complete it in logit space [see also Candès and Recht, 2009, Mazumder et al., 2010, Koren et al., 2009]. These methods use only each cell's aggregate score; Zeng and Papailiopoulos [2026] identify extending prediction to instance-level outcomes as a natural next step. PKPS unifies the two ends—a cell with $m \geq 0$ queries spans efficient benchmarking ($m>0$) and completion ($m=0$)—and adds the response-level signal, which can be ensembled with matrix completion.

3 The Product Kernel Perspective Space

Both paired DKPS and PKPS summarize the n models of Section 1.1 in an $n \times n$ affinity matrix A (defined below). The squared distance between two models is $D^2(a,b) = A_{aa} + A_{bb} - 2A_{ab}$ (clamped at 0), and their d-dimensional *representations* are the classical-multidimensional-scaling minimizer

$$(\hat{\psi}_1, \dots, \hat{\psi}_n) = \underset{z_1,\dots,z_n \in \mathbb{R}^d}{\arg\min} \sum_{i,k=1}^{n} \big(\|z_i - z_k\| - D(i,k)\big)^2, \tag{1}$$

so model i is the point $\hat{\psi}_i \in \mathbb{R}^d$, its *perspective*. Paired DKPS and PKPS differ only in the affinity A.

With all n models answering the same m queries ($Q_1 = \cdots = Q_n$), DKPS uses the identity-matched affinity $A_{ab} = \frac{1}{m}\sum_j k_R(x_{aj}, x_{bj})$: response j of model a is paired only with response j of model b, so the design must be paired.

PKPS replaces identity matching with a query kernel k_Q:

$$A_{ab} = \frac{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l)\, k_R(x_{aj}, x_{bl})}{\sum_{j \in Q_a} \sum_{l \in Q_b} k_Q(q_j, q_l)}, \tag{2}$$

where Q_a is the query set of model a. Each term is normalized by its own query-kernel mass, so models that answered different numbers of queries, or entirely different queries, remain comparable;

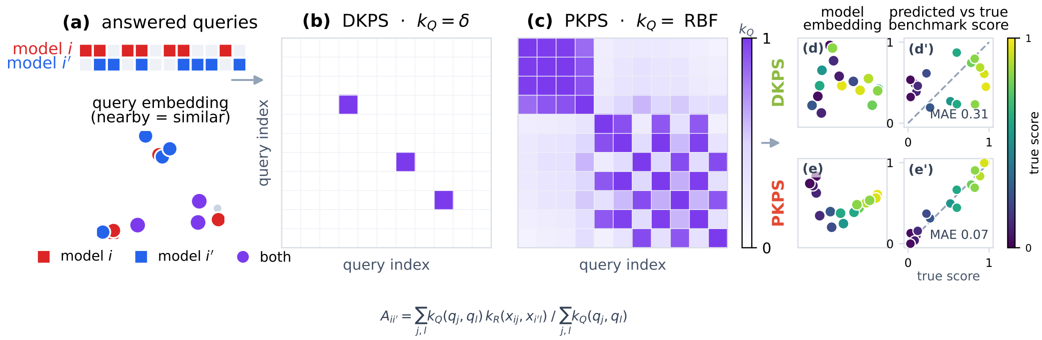

Figure 1: An illustration of the difference between the Product Kernel Perspective Space (PKPS) and the previously proposed Data Kernel Perspective Space (DKPS). **(a)** Top: two models respond to different sets of queries; bottom: queries represented as vectors. **(b)** The query kernel matrix used by DKPS: only information from queries both models have responded to is included in the dissimilarity matrix. **(c)** The query kernel matrix used by PKPS: information is weighted by query similarity in the embedding space. **(d–e′)** A toy benchmark-prediction example. Top: restricted to the rare shared queries, DKPS is noisy and correlates poorly with benchmark score (MAE 0.31). Bottom: PKPS leverages unpaired queries and is highly correlated with score (MAE 0.07).

responses to similar queries contribute even across tasks, so unpaired evaluations are used (Figure 1). Paired DKPS is the special case $k_Q = \delta$, where only $j = l$ survives.

We analyze and evaluate PKPS with an RBF query kernel over the embedded queries, $k_Q(q_j, q_l) = \exp(-\|u_j - u_l\|^2/2\sigma^2)$, with bandwidth σ chosen per run (below), and a linear response kernel $k_R(x, x') = \langle x, x' \rangle$; paired DKPS corresponds to a linear k_R with the identity query kernel $k_Q = \delta$. Linearity of k_R makes the kernel factorize across response domains (Section 4.1); the synthetic study uses an RBF k_R. Since Eq. 2 normalizes each pair by its own kernel mass, A need not be positive semidefinite; classical MDS embeds the doubly-centered matrix by its top-d nonnegative eigenvalues, as is standard. A_{ab} costs $|Q_a||Q_b|$ kernel terms per pair, and one response-kernel pass serves the whole bandwidth grid.

3.1 Theoretical properties of the PKPS

PKPS is a strict generalization of DKPS. With $k_Q = \delta$ only equal queries lying in both query sets survive the double sum of Eq. 2, so it reduces to paired DKPS over the shared queries, $A_{ab} = \frac{1}{|Q_a \cap Q_b|} \sum_{q \in Q_a \cap Q_b} k_R(x_{aq}, x_{bq})$; since the RBF kernel converges pointwise to δ as $\sigma \to 0$, DKPS is also the small-bandwidth limit. Conversely, for any strictly positive k_Q the normalizer is positive for any nonempty query sets, so A_{ab} is defined even on disjoint sets, where paired DKPS is undefined. Between these limits, the bandwidth σ controls the tradeoff between exact matching and bridging.

3.2 Inference in the PKPS

Inference is a two-step statistical problem, exactly as in the DKPS [Helm et al., 2026]: embed each model into the perspective space by the scaling of Eq. 1, then regress its benchmark score on the resulting coordinate. On reference models with known scores $\{(f_i, y_i)\}$ we fit $\hat{h} : \mathbb{R}^d \to \mathcal{Y}$ on the pairs $\{(\hat{\psi}_i, y_i)\}$ and predict $\hat{y}(f) = \hat{h}(\hat{\psi}(f))$. The free choices are the bandwidth σ and the regressor $\hat{h}$.

The optimal σ depends on the degree of pairing in the data. We choose it per run by cross-validation over a median-scaled grid $\sigma \in \{0.03, \ldots, 3\} \cdot \text{med}$, selecting the σ whose perspective best predicts the *observed* scores under leave-one-model-out k-NN. Note that the selection criterion uses only observed scores and never the held-out scores we report. Because the grid includes near-zero bandwidths, cross-validation recovers paired DKPS when query overlap is high.

For each replicate we (i) PCA-reduce response and query embeddings on the *observed* rows only (dimension by the second Zhu–Ghodsi elbow); (ii) form D from Eq. 2; (iii) embed models by classical MDS; (iv) regress scores on the embedding, fit leave-one-family-out. The score head—logit transform, per-task standardization, two-way (model + task) bias—is identical to the low-rank matrix-completion (LRMC) baseline's. The two methods differ only in how the residual is modeled (k-NN on the perspective versus a low-rank factorization), so performance differences are attributable to the representation. We found matching the per-task standardization to be important: without it, the model offset is dominated by the tasks with the largest logit variance.

The ensemble combines the PKPS estimate with whichever score signal is available for a cell. On an *observed* cell, the score signal is the cell's own sample score, and the blend weight is a scalar fit by the same leave-one-family-out cross-validation as the regressor, against the reference models' cached full scores; the fitted weight grows with the budget, so the sample score dominates as queries accumulate. On a *missing* cell, the score signal is LRMC [Zeng and Papailiopoulos, 2026], blended with PKPS by a weight cross-validated on observed scores. No weight uses the scores of the family being predicted. (A closed-form inverse-variance weighting performs slightly worse at every budget; Appendix D.)

3.3 Query efficiency

DKPS requires query *sets* to overlap; PKPS requires only query *distributions* to overlap. In this section we make this distinction precise by extending the machinery of Helm et al. [2026].

Definition 1 (Population PKPS)**.** Let each model a's (query, response) pairs be drawn i.i.d. from a design distribution P_a. The *population affinity* is $A^*_{ab} = \mathbb{E}[k_Q(u,u')\,k_R(x_a,x_b)]\,/\,\mathbb{E}[k_Q(u,u')]$ with draws from P_a and P_b independent, the population squared distances are $D^{*2}(a,b) = A^*_{aa} + A^*_{bb} - 2A^*_{ab}$, and the *population perspectives* $\psi_1,\dots,\psi_n \in \mathbb{R}^d$ are the rank-d spectral embedding of the centered Gram matrix B^* of D^{*2}.

Assumption 1 (Score regularity)**.** The benchmark score is bounded and γ-Lipschitz on the population perspective space.

In practice, Assumption 1 holds when models with similar response distributions attain similar scores; this is the working premise of any response-based predictor, and is visible as the smooth score gradient in Figure 1(e).

Assumption 2 (Model support)**.** The model distribution places mass on every neighborhood of the population perspective space.

Assumption 2 ensures that a target model has arbitrarily near reference neighbors once the cohort is large; in practice it suggests that a large, diverse reference population may be needed. Both conditions mirror Helm et al. [2026].

Assumption 3 (Kernel and geometry regularity)**.** Kernels and embedded responses are bounded; the *cross-model query-kernel mass* is bounded below, $\kappa = \inf_{a,b} \mathbb{E}[k_Q(u,u')] > 0$; and B^* has rank d with spectral gap $\lambda_d > 0$.

Boundedness and the spectral gap are standard conditions in spectral-embedding analyses. The important condition for the unpaired setting is $\kappa > 0$, the population analogue of the pairing fraction ρ: for an RBF k_Q it holds whenever the query *distributions* overlap, even when no individual query is shared. It fails when the query distributions are disjoint, i.e., when models are evaluated on domains that the kernel cannot bridge; in this case no unpaired method should be expected to compare them.

Proposition 1 (Concentration of PKPS perspectives)**.** *Under Assumption 3, with at least m queries per model, for any* $\delta \in (0,1)$*, with probability at least* $1-\delta$ *there is an orthogonal* W *such that* $\max_i \|\hat{\psi}_i - W\psi_i\| \le C\kappa^{-2}\lambda_d^{-1/2}\sqrt{d}\,n\sqrt{\log(4n^2/\delta)/2m}$*, where* C *depends only on the kernel bounds (proof in Appendix E).*

Definition 2 (Query efficiency; following Helm et al., 2026)**.** Fix a per-model budget m and let h_n denote a score estimator using n scored reference models. h is *query-efficient relative to* h' *at budget* m if there exists N such that $\mathrm{MSE}(h_n) \le \mathrm{MSE}(h'_n)$ with high probability for all $n \ge N$, and *query-efficient relative to* h' if this holds at every budget at which $\mathrm{MSE}(h'_n)$ is bounded away from zero.

Theorem 1 (Query efficiency of PKPS)**.** *Fix a budget* m *and pairing fraction* $\rho \in [0,1]$*, and let PKPS-CV select its bandwidth from a finite grid* Σ *containing the exact-matching limit* $k_Q = \delta$ *by validation on held-out reference scores. Under Assumptions 1–3, for any* $\varepsilon > 0$*:* (i) *there exist* (n,m)*, with* m *polynomial in* $(1/\varepsilon, 1/\kappa, n)$ *and independent of* ρ*, at which* $\mathrm{MSE}(\textit{PKPS-CV}) \le \varepsilon$ *with high probability;* (ii) *under a fully unpaired design (nonatomic design distributions) with non-degenerate scores,* $v = \mathrm{Var}(y) > 0$*, every paired-DKPS estimator has* $\mathrm{MSE} \ge v$ *at* every (n,m)*, so PKPS-CV is query-efficient relative to paired DKPS (Definition 2) with margin at least* $v - 2\varepsilon$*;* (iii) *at every pairing fraction,* $\mathrm{MSE}(\textit{PKPS-CV}) \le \mathrm{MSE}(\mathrm{DKPS}) + \varepsilon$ *with high probability for sufficiently many reference models, strictly whenever some bandwidth improves on exact matching.*

Corollary 1 (Relative to the sample score)**.** *The sample score's MSE at budget* m *is* σ_q^2/m *for every* n*, so by Theorem 1(i), PKPS-CV is query-efficient relative to the sample score whenever* $\varepsilon < \sigma_q^2/m$*, at every pairing fraction.*

Proof sketch. For (i), Proposition 1 bounds every perspective's estimation error; the nearest-neighbor error splits into two estimation terms and the population nearest-neighbor distance, which Assumption 2 makes small for n large; choosing $m = \mathrm{poly}(n, 1/\kappa, 1/\varepsilon)$ gives $\mathrm{MSE} \le \varepsilon$, and no step involves ρ—exact matching, by contrast, retains only the ρM_{ij} shared pairs, so the same argument degrades as $\rho \to 0$. For (ii), with no shared queries every identity-matched affinity is an empty sum, so a paired-DKPS prediction is a function of the reference data alone and its error is at least the score variance; (i) supplies a bandwidth attaining $\varepsilon < v$. (iii) is model selection: the δ member of the grid coincides with paired DKPS on the shared queries with the same regression head, and validation MSEs concentrate uniformly over the finite grid (Hoeffding), giving an oracle inequality. Full proofs in Appendix E. □

Theorem 1 states that a query budget independent of the pairing fraction suffices for any target accuracy (i); that under fully unpaired designs the separation from paired DKPS is information-theoretic—with no shared queries, a paired-DKPS prediction cannot depend on the target's responses at all (ii); and that cross-validation extends the comparison safely to every pairing fraction (iii). Only (iii) is a model-selection guarantee: (i) rests on the concentration of the self-normalized product-kernel affinity, in which the population overlap κ replaces the pairing fraction ρ (Proposition 1), and (ii) is an impossibility result for exact matching. In the synthetic experiments below, a bridging bandwidth improves on exact matching at every $\rho < 1$ (Figure 2e), so the strict case of (iii) is typical.

4 Experiments

We evaluate PKPS on synthetic data, where every lever of the observation process can be swept, and on two heterogeneous benchmark suites. Throughout, the two prediction problems of Section 1.1 structure the experiments: denoising observed cells (query efficiency) and predicting missing ones (completion).

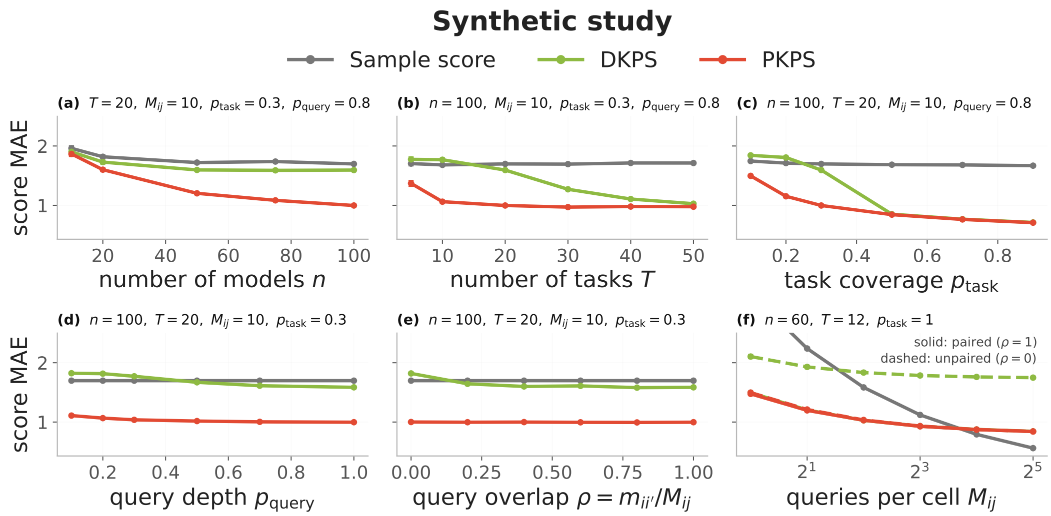

Figure 2: Score MAE on the synthetic benchmark as each lever of the observation process is varied (± 1 SEM over 50 seeds; DKPS green, PKPS red, sample score gray; fixed parameters in each panel title). **(a–e)** Completion of held-out cells as a function of cohort size n, number of tasks T, task coverage p_{task}, query depth p_{query}, and cross-model overlap $\rho = m_{ii'}/M_{ij}$; in (e), only DKPS degrades as $\rho \to 0$. **(f)** Denoising of observed cells: each cell is measured with M_{ij} noisy queries and we predict its true score, at paired (ρ=1, solid) and unpaired (ρ=0, dashed) designs; PKPS improves on the sample score at both, DKPS only when paired.

4.1 Evaluation setup

Synthetic data. We generate n models with latent ability vectors v_i and T tasks with latent vectors t_k, scores $v_i \cdot t_k + \varepsilon$, and queries drawn near each task center. Each model answers M_{ij} queries per task, of which $m_{ii'}$ are shared with another model (overlap $\rho = m_{ii'}/M_{ij}$); a task is observed with probability p_{task} and a query with probability p_{query}.

Benchmark suites. The first suite is 18 tasks across four HELM datasets with very different response types—MATH (7 math subjects), WMT-14 (5 translation directions), MedQA (medical multiple-choice), and LegalBench (5 legal classification subsets)—on the 93 models evaluated on all four. MATH and WMT have rich free-text responses; MedQA and LegalBench responses are single tokens for which text embeddings are degenerate, so we encode each dataset's responses natively (Gemini embeddings vs. one-hot answers) in disjoint blocks. With a linear k_R this makes cross-dataset response similarity exactly zero, and a single shared query embedding with a within-domain bandwidth keeps k_Q near zero across domains: the product kernel factorizes by domain, never comparing a proof to a legal label. The second suite, from the Every Eval Ever datastore [EvalEval Coalition, 2026], is 45 modern models on 16 tasks (MATH-MC levels 1–5, GSM-MC, GPQA-Diamond, and nine judging categories from JudgeBench and RewardBench 2), with per-cell query pools drawn independently per model ($\sim$10% incidental overlap)—unpaired by construction (Appendix B).

Prediction methods. We compare six predictors. The *sample score* is a cell's own m-query average. *IRT* fits a one-parameter logistic model on the largest shared item block [Polo et al., 2024]. *Low-rank matrix completion* (LRMC) is the logit-space imputation of Zeng and Papailiopoulos [2026] (missing cells only). *DKPS* and *PKPS* regress scores on the model embeddings of Section 3, DKPS being the identity-query-kernel limit. The *ensemble* combines PKPS with whichever score signal a cell has (Section 3).

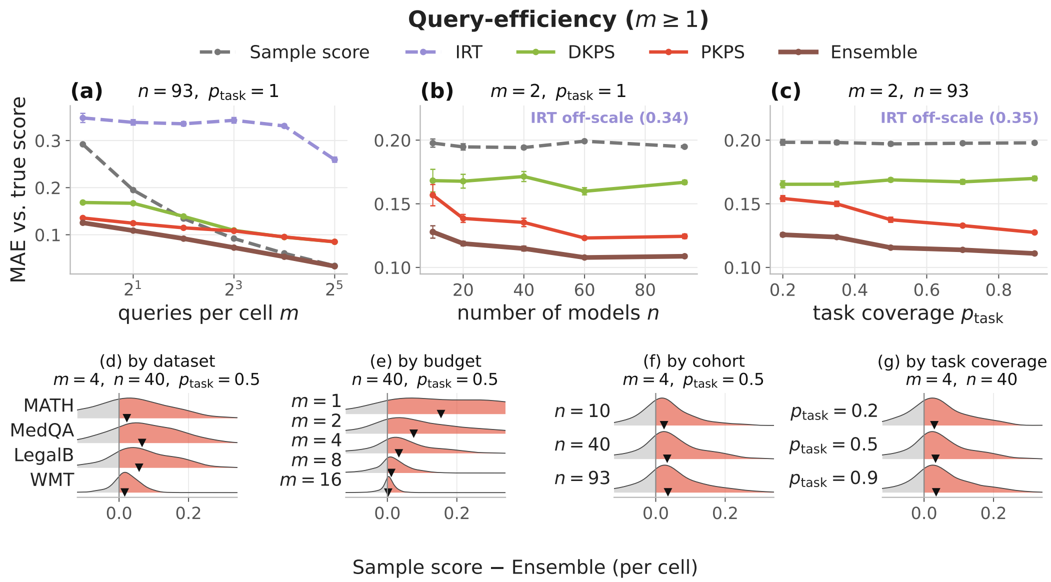

Figure 3: Query-efficient evaluation on the HELM suite. **(a–c)** MAE against the true full-evaluation score when each model answers m unpaired queries per observed cell (± 1 SEM), as a function of budget m, cohort size n, and task coverage p_{task}. IRT is poorly identified without a shared item block and is omitted from (b, c); its value is annotated. **(d–g)** Per-cell difference $\Delta =$ Sample score $-$ Ensemble MAE (red = Ensemble better) by dataset, budget, cohort, and coverage, with the other levers fixed at their sweep median; ▼ marks the per-row mean.

Measurement. Every experiment predicts held-out or true cell scores and reports MAE against the full-evaluation score, mean ± 1 SEM over 16 seeds (50 on synthetic data). Sweeps vary one observation lever at a time, holding the others fixed (grids in Appendix A); all hyperparameters, including the query bandwidth, are chosen per run by the leak-free cross-validation of Section 3.

4.2 Results

PKPS outperforms DKPS on every observation lever. Figure 2 shows score MAE for PKPS, DKPS, and the sample score as each lever of the observation process is varied. PKPS outperforms DKPS at every setting, and both outperform the sample score. The margin is wide even when the data is mostly paired (a–c), since bridging similar queries, across tasks as well as within, supplies signal that exact matching does not use. The gap is largest when pairing is scarce: as p_{query} drops (d), DKPS degrades while PKPS is stable, and when ρ is swept directly (e), DKPS approaches the sample score as $\rho \to 0$ while PKPS is unaffected.

Panel (f) considers the *denoising* form of the same problem: each cell is now *observed* with M_{ij} noisy queries and we predict its true score. PKPS with a single query matches the accuracy that the sample score reaches at roughly five queries, whether or not the queries are paired; DKPS denoises only when paired (dashed). At large budgets the sample score becomes precise enough to overtake the embedding's residual bias. This is the empirical counterpart of Theorem 1.

PKPS-based methods outperform the sample score at small query budgets. Figure 3 shows MAE as a function of the query budget m. Each model answers its own independently sampled queries (the real-data analogue of ρ=0; IRT is poorly identified in this regime) and we predict the true full-evaluation score. At small budgets the embedding-based methods are substantially more accurate than the sample score: at m=1, PKPS attains 0.136 and the ensemble 0.125 versus 0.292, so PKPS with one query matches the sample score at four. As the budget grows, the sample score

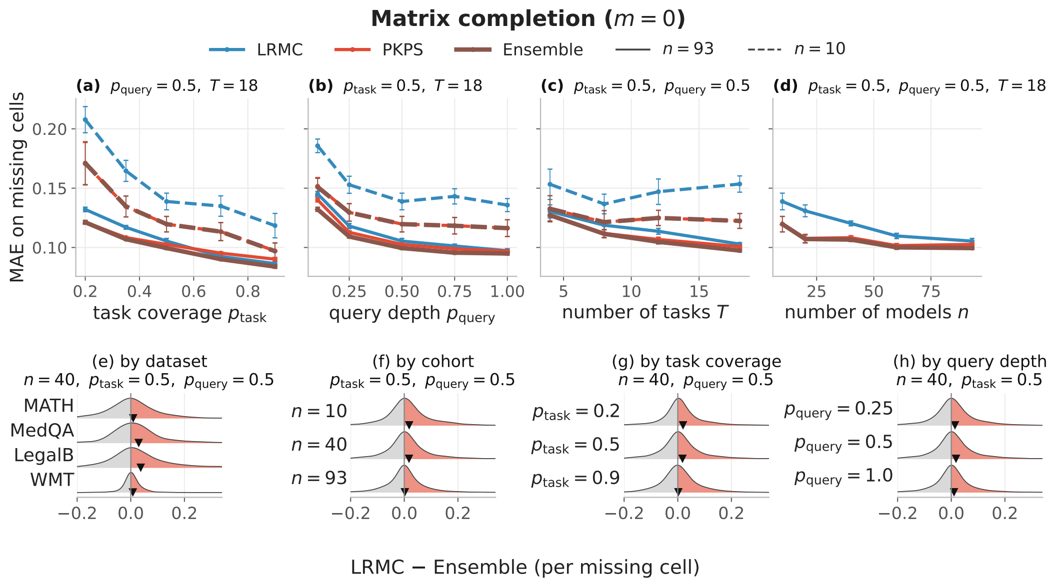

Figure 4: Matrix completion on the HELM suite. **(a–d)** MAE on missing cells (± 1 SEM) when a fraction p_{task} of (model, task) cells is observed at query depth p_{query}, as a function of task coverage, query depth, number of tasks, and cohort size (LRMC blue, PKPS red, Ensemble brown; solid n=93, dashed n=10). **(e–h)** Per-missing-cell difference Δ = LRMC − Ensemble MAE (red = Ensemble better) by dataset, cohort, coverage, and query depth, with the other levers fixed at their sweep median; ▼ marks the per-row mean.

eventually overtakes the embedding's residual bias (0.034 vs. 0.085 at m=32); the ensemble follows the better of its two components throughout. Panels (b, c) vary the cohort and coverage: PKPS improves steadily (0.157→0.124, 0.154→0.128; 35–37% on the second suite) while DKPS, which cannot use the additional unpaired responses, is flat.

The embedding completes missing cells where LRMC cannot. The same embedding also predicts cells that were *never run*. Figure 4 shows MAE on missing cells when a fraction p_{task} of cells is observed (each at depth p_{query}), against the LRMC baseline (the sample score is unavailable for missing cells). LRMC requires many rows to estimate its factors and degrades when models are scarce: at n=10 its MAE is 0.139 versus 0.120 for PKPS (Figure 4, dashed). At the full cohort (n=93, solid) PKPS slightly edges LRMC (0.103 vs. 0.105) and the ensemble performs best (0.099).

The per-cell gains are broad but tail-driven. The Ensemble outperforms the sample score on most cells, with the largest gains at the smallest budgets and on the rich-response datasets (Figure 3d–g); the win over LRMC is modest but stays positive in the mean across datasets, cohorts, coverage, and query depth (Figure 4e–h).

Both applications replicate on the Every Eval Ever suite. The second suite shows the same qualitative behavior with a larger completion margin (Table 1; lever sweeps in Appendix B): at m=1 PKPS attains a third of the sample score's error (0.076 vs. 0.224), and on missing cells the ensemble outperforms LRMC at *every* operating point, including the full cohort (0.069 vs. 0.096 at n=45). Table 1 collects both applications on both suites; the ensemble is best or tied in every column.

The results are insensitive to representation and pipeline choices. Varying the MDS dimension over 4–24 moves PKPS's MAE by less than 0.005, and the cross-validated bandwidth matches the best fixed choice (Appendix C). Ablations of the regression head, the response kernel, the ensemble weighting rule, and the incidental query overlap in the EEE pools are in Appendix D.

5 Discussion

In this paper we introduced the PKPS, which compares models through their responses to similar, rather than identical, queries. Outside of curated leaderboards, evaluation data is typically unpaired: each model answers the queries its evaluators chose, and methods that require a shared block of identical queries do not apply. Our central finding is that benchmark prediction remains possible under unpaired evaluation: as $\rho \to 0$, DKPS and IRT degrade sharply while PKPS does not. The same embedding supports both applications we studied, denoising cheap evaluations and completing never-run cells, and the ensemble performs at least as well as either channel alone.

Limitations and future work. We close with the current limitations and the extensions they suggest. First, PKPS embeds every observed response and query with a single fixed embedder; we did not ablate this choice, and embedding functions are known to have a nonnegligible effect at small budgets [Helm et al., 2026]. Second, the block-diagonal instantiation on the HELM suite disables bridging across response domains; a shared response space that would let responses in one domain inform another remains untested. Third, the affinity is quadratic in the number of queries per model pair, though one response-kernel pass serves the whole bandwidth grid. Fourth, Theorem 1(i) is existential (it provides no rate in n), and the strict margin of (iii) at intermediate ρ rests on a bandwidth improving on exact matching, which we establish empirically rather than by a lower bound. Finally, per-cell gains over score-only baselines are modest on dense low-rank matrices; they are decisive in the sparse, noisy, unpaired regime that motivates the problem.

Reproducibility statement

All methods are released as a documented Python package (the PKPS estimator, matrix completion, and IRT baselines), with the data-preparation, experiment, and figure scripts for both the synthetic study and the HELM suite; every experiment is seeded, all real-data numbers are means over 16 seeds (synthetic: 50), and hyperparameters are selected by the leak-free cross-validation described in Section 3 (leakage is checked by an automated test that corrupts one model's held-out scores and verifies its predictions are unchanged). Appendix A lists the suite composition, embedding models, and all hyperparameter settings.

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A Experimental details

Suite. 18 tasks over four datasets from HELM [Liang et al., 2023]: MATH (7 subjects), WMT-14 (5 translation directions), MedQA, and LegalBench (5 subsets), on the 93 models evaluated on all four; Table 2 lists every task with its query count (the loader subsamples each task to at most 120 queries with a fixed seed, so datasets contribute comparably). MATH and WMT-14 responses are free text and are embedded with `gemini-embedding-001` (3072-d); MedQA and LegalBench responses are single-token answers, encoded one-hot. Queries share a single `gemini-embedding-001` embedding space. Response and query embeddings are reduced by PCA fit on observed rows only, with dimension chosen by the second Zhu–Ghodsi elbow.

Model coverage (HELM). The 93 shared models span 20 developers: `01-ai_yi-34b`, `01-ai_yi-6b`, `01-ai_yi-large-preview`, `AlephAlpha_luminous-base`, `AlephAlpha_luminous-extended`, `AlephAlpha_luminous-supreme`, `ai21_j2-grande`, `ai21_j2-jumbo`, `ai21_jamba-1.5-large`, `ai21_jamba-1.5-mini`, `ai21_jamba-instruct`, `allenai_olmo-7b`, `amazon_nova-lite-v1`, `amazon_nova-micro-v1`, `amazon_nova-pro-v1`, `anthropic_claude-2.0`, `anthropic_claude-2.1`, `anthropic_claude-3-5-haiku-20241022`, `anthropic_claude-3-5-sonnet-20240620`,

Table 2: HELM suite composition: 18 tasks over four datasets, with the number of unique queries per task. Every task is scored for all 93 shared models; the loader caps each task at 120 queries (seeded subsample).

Dataset	Task	Queries	Dataset	Task	Queries
MATH	algebra	135	WMT-14	cs–en	873
MATH	counting & probability	39	WMT-14	de–en	776
MATH	geometry	38	WMT-14	fr–en	754
MATH	intermediate algebra	52	WMT-14	hi–en	387
MATH	number theory	30	WMT-14	ru–en	613
MATH	prealgebra	86	MedQA	med_qa	1000
MATH	precalculus	57	LegalBench	abercrombie	95
LegalBench	corporate lobbying	490	LegalBench	citizenship questions	1000
LegalBench	function of decision section	367	LegalBench	proa	95

`anthropic_claude-3-5-sonnet-20241022`, `anthropic_claude-3-haiku-20240307`, `anthropic_claude-3-opus-20240229`, `anthropic_claude-3-sonnet-20240229`, `anthropic_claude-instant-1.2`, `anthropic_claude-instant-v1`, `anthropic_claude-v1.3`, `cohere_command`, `cohere_command-light`, `cohere_command-r`, `cohere_command-r-plus`, `databricks_dbrx-instruct`, `deepseek-ai_deepseek-llm-67b-chat`, `deepseek-ai_deepseek-v3`, `google_gemini-1.0-pro-001`, `google_gemini-1.0-pro-002`, `google_gemini-1.5-flash-001`, `google_gemini-1.5-flash-002`, `google_gemini-1.5-pro-001`, `google_gemini-1.5-pro-002`, `google_gemini-1.5-pro-preview-0409`, `google_gemini-2.0-flash-exp`, `google_gemma-2-27b-it`, `google_gemma-2-9b-it`, `google_gemma-7b`, `google_text-bison@001`, `google_text-unicorn@001`, `meta_llama-2-13b`, `meta_llama-2-70b`, `meta_llama-2-7b`, `meta_llama-3-70b`, `meta_llama-3-8b`, `meta_llama-3.1-405b-instruct-turbo`, `meta_llama-3.1-70b-instruct-turbo`, `meta_llama-3.1-8b-instruct-turbo`, `meta_llama-3.2-11b-vision-instruct-turbo`, `meta_llama-3.2-90b-vision-instruct-turbo`, `meta_llama-3.3-70b-instruct-turbo`, `meta_llama-65b`, `microsoft_phi-2`, `microsoft_phi-3-medium-4k-instruct`, `microsoft_phi-3-small-8k-instruct`, `mistralai_mistral-7b-instruct-v0.3`, `mistralai_mistral-7b-v0.1`, `mistralai_mistral-large-2407`, `mistralai_mistral-medium-2312`, `mistralai_mixtral-8x22b`, `mistralai_mixtral-8x7b-32kseqlen`, `mistralai_open-mistral-nemo-2407`, `nvidia_nemotron-4-340b-instruct`, `openai_gpt-3.5-turbo-0613`, `openai_gpt-4-0613`, `openai_gpt-4-1106-preview`, `openai_gpt-4-turbo-2024-04-09`, `openai_gpt-4o-2024-05-13`, `openai_gpt-4o-2024-08-06`, `openai_gpt-4o-mini-2024-07-18`, `openai_text-davinci-002`, `openai_text-davinci-003`, `qwen_qwen1.5-110b-chat`, `qwen_qwen1.5-14b`, `qwen_qwen1.5-32b`, `qwen_qwen1.5-72b`, `qwen_qwen1.5-7b`, `qwen_qwen2-72b-instruct`, `qwen_qwen2.5-72b-instruct-turbo`, `qwen_qwen2.5-7b-instruct-turbo`, `snowflake_snowflake-arctic-instruct`, `tiiuae_falcon-40b`, `tiiuae_falcon-7b`, `upstage_solar-pro-241126`, `writer_palmyra-x-004`, `writer_palmyra-x-v2`, `writer_palmyra-x-v3`.

PKPS. RBF query kernel with bandwidth selected per run by leave-one-model-out k-NN (k=8) on the observed scores over the grid $\sigma \in \{0.03, 0.1, 0.3, 1, 3\} \times$ median pairwise query distance, plus the near-delta (DKPS) limit; linear response kernel on real data, RBF on synthetic. Response blocks are capped at 48 PCA components per dataset and the shared query space at 64; each (model, query) pair uses one sampled response (r=1). Models are embedded by classical MDS with d=8 throughout, following Helm et al. [2026] and fixed a priori (d = latent dimension on synthetic). Appendix C shows the results are insensitive to d over 4–24, and that the cross-validated bandwidth tracks the best fixed choice.

Prediction head. All embedding methods share one head: logit-transformed scores, per-task standardization to unit variance, a two-way (model + task) additive bias, and a distance-weighted k-NN (k=5) regression of the residual on the model embedding, fit *leave-one-family-out*: all models from the target's developer are excluded from its training set, so predictions never lean on near-duplicate siblings [Helm et al., 2026]. Held-out cells are predicted directly; observed cells leave their own target out.

Baselines. Low-rank matrix completion (LRMC) follows BenchPress [Zeng and Papailiopoulos, 2026]: logit space, per-task standardization, two-way bias, rank-2 ALS with ridge λ=0.1, averaged over 5 random initializations; on observed cells it is cross-fit so a cell's prediction never sees its own sample. IRT is a 1PL (Rasch) model on binary tasks: item difficulties by maximum likelihood (L-BFGS) on the largest fully-observed (model $\times$ item) block, per-model ability by bounded scalar MLE, clipped to $[-4,4]$. The sample baseline is the cell's own m-query mean.

Ensemble. On observed cells, the prediction is $\alpha\,\hat{y}_{\text{sample}} + (1-\alpha)\,\hat{y}_{PKPS}$ with a scalar α per held-out family, selected on a 21-point grid by leave-one-family-out validation against the other families' cached full scores (the same information the regressor trains on); on missing cells, PKPS and LRMC are blended with a weight fit by held-out cross-validation against observed scores. No quantity ever uses the scores of the family being predicted. Appendix D compares this rule to a closed-form inverse-variance weighting and to the fixed m/M schedule of Helm et al. [2026].

Sweeps and reporting. All real-data sweeps use 16 seeds; curves and table entries report the per-seed mean over tasks, then the mean ($\pm$1 SEM) over seeds. Query efficiency: budgets $m \in \{1,2,4,8,16,32\}$ at full cohort and coverage; cohort $n \in \{10,20,40,60,93\}$ (HELM) or $\{10,20,30,45\}$ (EEE) and task coverage $p_{\text{task}} \in \{0.2,0.35,0.5,0.7,0.9\}$, both at the few-query point m=2. Completion: task coverage as above and query depth $p_{\text{query}} \in \{0.1,0.25,0.5,0.75,1.0\}$ with cohort lines $n \in \{10,\text{full}\}$; task count $T \in \{4,8,12,18\}$ (HELM) or $\{4,8,12,16\}$ (EEE); the cohort sweep holds p_{task}=p_{query}=0.5.

Synthetic study. 50 seeds per configuration; latent dimension 5, observed embedding dimension 20, score noise 0.1, response noise 0.5; both kernels RBF; k-NN regression with k=5. Panel-specific settings are given in each figure title.

Compute. All experiments run on a single multi-core CPU node; text embeddings are precomputed once through the embedding API and cached. Embedding the EEE suite (30,304 pooled responses plus 7,665 queries, $\sim$11M tokens) took $\sim$13 minutes and under \$2 of API credit; each 16-seed sweep completes in minutes (EEE) to hours (HELM) on the same node.

B The Every Eval Ever suite

The second suite is built from the Every Eval Ever datastore [EvalEval Coalition, 2026], an MIT-licensed, community-maintained collection of item-level evaluation results in a unified format (inputs, raw model outputs, and per-item scores). We take the five benchmarks with the widest item-level model coverage—MATH-MC, GSM-MC, GPQA-Diamond, JudgeBench, and RewardBench 2—restricted to the 45 models present on all five; sub-tasks (MATH-MC levels 1–5; four JudgeBench and five RewardBench 2 categories) give 16 tasks (Table 3). Every domain is free text (including the judging benchmarks' rationales), so all responses and queries are embedded with `gemini-embedding-001` and blocked by benchmark exactly as in Appendix A. Each cell's observable pool is 32 queries drawn with a model-dependent seed, so two models share only incidental queries ($\sim$10%); the query-efficiency budget m and the completion depth p_{query} subsample this pool, and the prediction target is always the cell's full-depth mean over all items, which is never embedded. All scores are binary, so IRT applies to every task (on the HELM suite it excludes WMT). Figures 5 and 6 repeat the lever sweeps of Figures 3 and 4 on this suite; Table 1 summarizes.

Model coverage (EEE). The 45 models shared by all five benchmarks span 13 developers:
`cohere/c4ai-command-a-03-2025, cohere/c4ai-command-r-08-2024,`
`cohere/c4ai-command-r-plus-08-2024, cohere/c4ai-command-r7b-12-2024,`
`cohere/command-a-reasoning-08-2025, deepseek/deepseek-r1-0528,`
`deepseek/deepseek-v3-1-terminus, deepseek/deepseek-v3-2, deepseek/deepseek-v3-2-speciale,`
`deepseek/deepseek-v4-flash-fp8, deepseek/deepseek-v4-pro, google/gemini-3-1-pro-preview,`

Table 3: EEE suite composition: 16 tasks over five benchmarks, with the number of items per (model, task) cell. Every task is fully answered by all 45 shared models; each cell's observable pool is 32 items, sampled per model.

Benchmark	Task	Items	Benchmark	Task	Items
MATH-MC	level 1	430	JudgeBench	coding	42
MATH-MC	level 2	882	JudgeBench	knowledge	154
MATH-MC	level 3	1115	JudgeBench	math	56
MATH-MC	level 4	1199	JudgeBench	reasoning	98
MATH-MC	level 5	1288	RewardBench 2	factuality	475
GSM-MC	gsm_mc	1319	RewardBench 2	focus	495
GPQA-Diamond	gpqa_diamond	198	RewardBench 2	math	183
RewardBench 2	precise if	160	RewardBench 2	safety	450

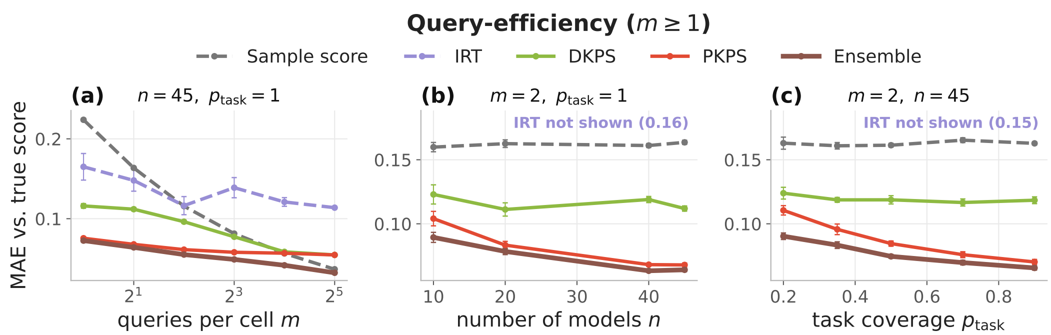

Figure 5: Query-efficient evaluation on the Every Eval Ever suite (45 models, 16 tasks; MAE against the true full-evaluation score, ± 1 SEM over 16 seeds). The protocol matches Figure 3. PKPS and the Ensemble are the most accurate at every budget; DKPS trails because the query pools are unpaired, and IRT is poorly identified.

`google/gemma-2-27b-it, google/gemma-2-9b-it, google/gemma-3-12b-it, google/gemma-3-27b-it, google/gemma-4-e2b-it, google/gemma-4-e4b-it, llm360/k2-v2-instruct, meta/llama-4-maverick-17b-128e-instruct-fp8, meta/meta-llama-3-1-8b-instruct, minimax/minimax-m2-5, mistral/mistral-small-4-119b-2603, moonshot/kimi-k2-5, moonshot/kimi-k2-6, moonshot/kimi-k2-thinking, openai/gpt-5-5, openai/gpt-5-mini, openai/gpt-5-nano, openai/gpt-oss-120b, openai/gpt-oss-20b, qwen/qwen3-30b-a3b-thinking-2507, qwen/qwen3-5-122b-a10b, qwen/qwen3-5-2b, qwen/qwen3-5-397b-a17b, qwen/qwen3-6-35b-a3b, qwen/qwen3-6-plus, qwen/qwen3-next-80b-a3b-thinking, stepfun/step-3-5-flash, xiaomi/mimo-v2-flash, zai-org/glm-4-6-fp8, zai-org/glm-4-7-flash, zai-org/glm-4-7-fp8, zai-org/glm-5-1-fp8, zai-org/glm-5-fp8.`

C Sensitivity analyses

Figure 7 probes the two representation hyperparameters on the query-efficiency experiment (16 seeds throughout). **MDS dimension (a, b).** Repeating the budget sweep at $d \in \{4, 8, 24\}$ moves PKPS's MAE by less than 0.005 at every budget on both suites (at the smallest budgets, smaller d is very slightly better—fewer coordinates to estimate from one noisy query per cell); the paper's choice (d=8, following Helm et al., 2026) is never the worst point at any budget. **Query bandwidth (c).** Fixing the bandwidth at each grid multiplier instead of cross-validating traces the expected curve: near-delta multipliers and the exact-matching δ limit are substantially worse (unpaired pools leave them little to compare), broad multipliers are best, and the leak-free cross-validated choice matches the best fixed multiplier on both suites (0.126 vs. 0.125 on HELM, 0.069 vs. 0.068 on EEE)—the one hyperparameter that matters is the one the method selects automatically. Sensitivity to the number of reference models is part of the main experiments (the cohort sweeps of Figures 3b and 4d).

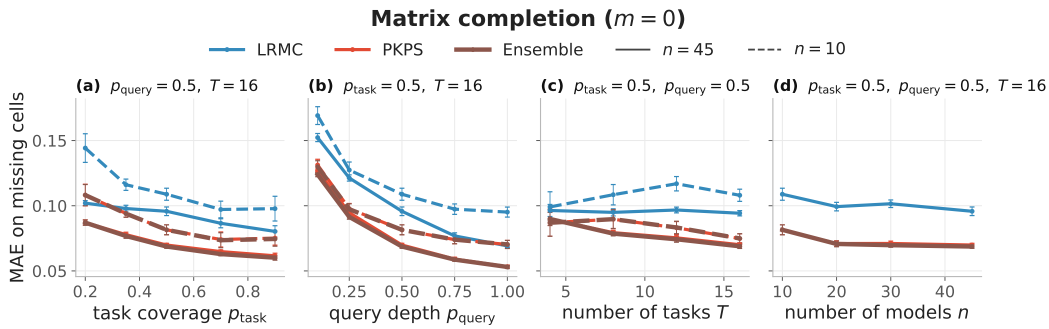

Figure 6: Matrix completion on the Every Eval Ever suite (MAE on missing cells, ± 1 SEM; solid n=45, dashed n=10). The protocol matches Figure 4. Unlike on the HELM suite, PKPS and the Ensemble outperform LRMC at every operating point.

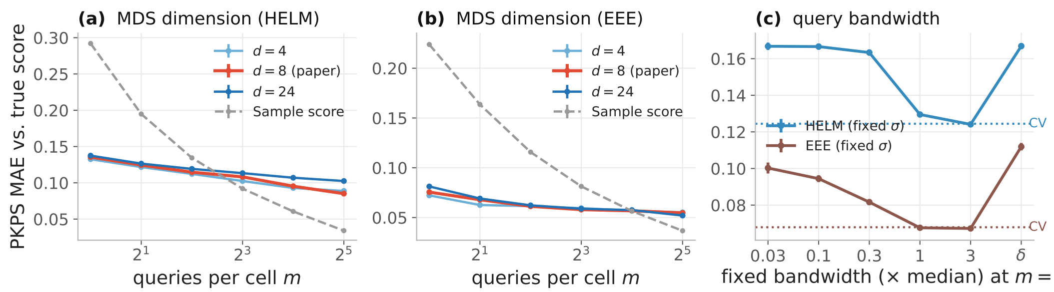

Figure 7: Sensitivity analyses on the query-efficiency experiment (± 1 SEM, 16 seeds). **(a, b)** PKPS MAE against budget at MDS dimension $d \in \{4, 8, 24\}$ (paper choice in red; sample dashed); the curves are nearly indistinguishable on both suites. **(c)** MAE at m=2 with the query bandwidth fixed at each grid multiplier (δ = exact matching); dotted lines mark the cross-validated selection, which matches the best fixed choice on both suites.

D Pipeline ablations

Each ablation repeats the query-efficiency budget sweep (16 seeds, both suites) with one pipeline choice changed; all other settings match Appendix A.

Ensemble weighting. We compare the paper's cross-validated scalar weight against two alternatives: a closed-form inverse-variance rule (the sample score weighted by $e^2_{PKPS}/(e^2_{PKPS} + \tilde{p}(1 - \tilde{p})/m)$ per cell, with $\tilde{p}$ the PKPS estimate and e^2_{PKPS} the depth-weighted, noise-debiased disagreement with the observed samples) and the fixed schedule $\alpha = m/M$ of Helm et al. [2026]. The cross-validated weight is best at every budget on both suites: on HELM it improves on the inverse-variance rule from 0.132 to 0.125 at m=1 and from 0.077 to 0.073 at m=8; on EEE from 0.084 to 0.073 and from 0.052 to 0.049. The m/M schedule trails both on HELM (0.096 at m=8) and is close to the cross-validated weight on EEE, where the small pools (M=32) let m/M reach 1.

Regression head. Replacing the distance-weighted k-NN head with ordinary least squares helps slightly on HELM (0.132 vs. 0.136 at m=1; 0.080 vs. 0.085 at m=32), consistent with Helm et al. [2026], but hurts on EEE, increasingly so at large budgets (0.066 vs. 0.055 at m=32). We use k-NN as the robust default across suites.

Response kernel. An RBF response kernel in place of the linear one changes PKPS's MAE by at most 0.003 at every budget on both suites; the linear choice is justified by the block-diagonal factorization (Section 4.1) at no accuracy cost.

Incidental overlap. The EEE pools share $\sim$10% of items across models by chance. Enforcing exactly disjoint query sets (true $\rho = 0$) changes PKPS by at most 0.005 at $m \le 8$ (0.080 vs. 0.076 at m=1), and IRT becomes inapplicable outright—no shared item block exists. At $m \ge 16$ the disjointness constraint exhausts the 32-item pools and mechanically shortens the sample score's effective depth, so comparisons there reflect the constraint rather than the methods.

E Proofs of Theorem 1 and Corollary 1

Population quantities. Queries for model a are drawn i.i.d. from a design distribution over embedded queries; write $z = (u, x_a(u))$ for an embedded query–response pair (response noise included) and P_a for its law. By Assumption 3 the kernels are bounded: $k_Q \in [0,1]$ after normalization (RBF) and $|k_R| \le B_R$. Define the population affinity

$$A^*_{ab} \;=\; \frac{\mathbb{E}\left[k_Q(u,u')\,k_R(x_a(u),x_b(u'))\right]}{\mathbb{E}\left[k_Q(u,u')\right]}, \qquad z \sim P_a,\; z' \sim P_b \text{ independent,}$$

population squared distances $D^{*2}(a,b) = A^*_{aa} + A^*_{bb} - 2A^*_{ab}$, the centered Gram matrix $B^* = -\frac{1}{2} J D^{*2} J$ with $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$, and population perspectives $\Psi^* \in \mathbb{R}^{n\times d}$ its rank-d spectral embedding, which exists with $\lambda_d(B^*) > 0$ by Assumption 3 (cf. Definition 1).

Lemma 1 (Affinity concentration)**.** *Under Assumption 3, for models a,b with $\min(|Q_a|,|Q_b|) \ge m$ and any $\delta \in (0,1)$: with probability at least $1-\delta$,*

$$|A_{ab} - A^*_{ab}| \;\le\; \frac{2(1+B_R)}{\kappa^2}\, t_m(\delta), \qquad t_m(\delta) = \bar{B}\sqrt{\frac{\log(4/\delta)}{2m}}, \quad \bar{B} = \max(2B_R, 1),$$

provided $t_m(\delta) \le \kappa/2$.

Proof. The numerator $N = \frac{1}{|Q_a||Q_b|}\sum_{j,l} k_Q(u_j,u'_l)\,k_R(x_{aj},x_{bl})$ and denominator $Z = \frac{1}{|Q_a||Q_b|}\sum_{j,l} k_Q(u_j,u'_l)$ of Eq. 2 are two-sample U-statistics of order $(1,1)$ in the i.i.d. pairs $z_j \sim P_a$, $z'_l \sim P_b$, with kernels of range at most $2B_R$ and 1 respectively (response noise enters only through the bounded kernel argument, so the number of responses per query may stay fixed). By Hoeffding's inequality for two-sample U-statistics [Hoeffding, 1963, Section 5], each deviates from its mean by more than its range times $\sqrt{\log(4/\delta)/2m}$ with probability at most $\delta/2$; since $\bar{B}$ dominates both ranges, a union bound gives $|N - N^*| \le t_m(\delta)$ and $|Z - Z^*| \le t_m(\delta)$ jointly with probability $\ge 1-\delta$. On this event, since $Z^* \ge \kappa$ by Assumption 3 and $t_m(\delta) \le \kappa/2$, we have $Z \ge \kappa/2$, and

$$|A_{ab} - A^*_{ab}| = \frac{|NZ^* - N^*Z|}{ZZ^*} \;\le\; \frac{|N-N^*|}{Z} + \frac{|N^*|\,|Z-Z^*|}{ZZ^*} \;\le\; \frac{2\,t_m(\delta)}{\kappa} + \frac{2B_R\,t_m(\delta)}{\kappa^2} \;\le\; \frac{2(1+B_R)}{\kappa^2}\,t_m(\delta),$$

using $|N^*| \le B_R$ and $\kappa \le 1$.

□

Lemma 2 (Perspective perturbation)**.** *Let $\varepsilon_\infty = \max_{a,b}|A_{ab} - A^*_{ab}|$. There is an orthogonal $W \in \mathbb{R}^{d\times d}$ such that*

$$\max_i \;\|\hat{\psi}_i - W\psi_i\| \;\le\; \frac{25\sqrt{d}\,n\,\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}\,.$$

Proof. Entrywise, $|D^2 - D^{*2}| \leq 4\varepsilon_\infty$, and double centering inflates an entrywise bound by at most a factor of 4, so $E = B - B^*$ satisfies $\|E\|_{\mathrm{op}} \leq \|E\|_F \leq 8\,n\,\varepsilon_\infty$. Let B_d be the top-d truncation of B that classical MDS embeds ($B_d = \hat{\Psi}\hat{\Psi}^\top$, positive semidefinite). By Weyl's inequality every eigenvalue of B beyond the d-th, and every negative eigenvalue, has magnitude at most $\|E\|_{\mathrm{op}}$ (since $\lambda_{d+1}(B^*) = 0$ and $B^* \succeq 0$), so $\|B_d - B\|_{\mathrm{op}} \leq \|E\|_{\mathrm{op}}$. The matrix $B_d - B^*$ has rank at most $2d$, hence

$$\|B_d - B^*\|_F \;\leq\; \sqrt{2d}\,\|B_d - B^*\|_{\mathrm{op}} \;\leq\; \sqrt{2d}\,\big(\|B_d - B\|_{\mathrm{op}} + \|E\|_{\mathrm{op}}\big) \;\leq\; 2\sqrt{2d}\,\|E\|_F.$$

Both $B_d = \hat{\Psi}\hat{\Psi}^\top$ and $B^* = \Psi^*\Psi^{*\top}$ are positive semidefinite of rank at most d, so by the Procrustes alignment lemma of Tu et al. [2016, Lemma 5.4],

$$\min_{W \in O(d)} \|\hat{\Psi} - \Psi^* W\|_F^2 \;\leq\; \frac{\|B_d - B^*\|_F^2}{2(\sqrt{2}-1)\,\lambda_d(B^*)}.$$

Combining, $\max_i \|\hat{\psi}_i - W\psi_i\| \leq \|\hat{\Psi} - \Psi^* W\|_F \leq \frac{2\sqrt{2d}}{\sqrt{2(\sqrt{2}-1)}} \cdot \frac{8n\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}} \leq \frac{25\sqrt{d}\,n\varepsilon_\infty}{\sqrt{\lambda_d(B^*)}}$. Nearest-neighbor distances are invariant to W. □

Proof of Theorem 1(i). Take scores in $[0,1]$ without loss of generality (Assumption 1). Fix $\varepsilon > 0$ and set targets $c \leq \sqrt{\varepsilon/2}/(8\gamma)$ for the estimation error and $\varepsilon' \leq \sqrt{\varepsilon/2}/(2\gamma)$ for the population nearest-neighbor distance. By Assumption 2 there is $n_0(\varepsilon')$ such that for $n \geq n_0$, a reference model within population distance ε' of the target exists except with probability at most $\varepsilon/8$. Fix such an n. Applying Lemma 1 with confidence $\delta = \varepsilon/(8n^2)$ to every pair, a union bound and Lemma 2 give $\max_i \|\hat{\psi}_i - W\psi_i\| \leq c$ except with probability at most $\varepsilon/8$, once

$$m \;\geq\; C'\,\frac{d\,(1+B_R)^2\,\bar{B}^2\,n^2\log(n/\varepsilon)}{\kappa^4\,\lambda_d\,c^2} \;=\; \mathrm{poly}(n, 1/\kappa, 1/\varepsilon),$$

for an absolute constant C' (this choice also satisfies the proviso $t_m \leq \kappa/2$ of Lemma 1), and does not involve the pairing fraction ρ. On the intersection of the two events, for a target model f with nearest estimated reference neighbor f^* (perspectives $\psi^*, \hat{\psi}^*$; the target's $\psi, \hat{\psi}$),

$$|\hat{y}(f) - y(f)| \;\leq\; \gamma\|\psi^* - \psi\| \;\leq\; \gamma\big(\|\hat{\psi}^* - W\psi^*\| + \|\hat{\psi}^* - \hat{\psi}\| + \|\hat{\psi} - W\psi\|\big) \;\leq\; \gamma(2c + \hat{\delta}),$$

using Assumption 1 and orthogonal invariance. The estimated nearest-neighbor distance $\hat{\delta}$ is at most the estimated distance to the ε'-close reference, itself at most $\varepsilon' + 2c$ by the same alignment argument, so $|\hat{y} - y| \leq \gamma(4c + \varepsilon') \leq \sqrt{\varepsilon/2}$. Squaring, and adding the two failure events (squared errors are bounded by 1), $\mathrm{MSE}(\hat{y}) \leq \varepsilon/2 + \varepsilon/4 \leq \varepsilon$. □

Relative to the sample score. The sample score at budget m is the mean of m i.i.d. per-query scores and uses no reference models, so its MSE is σ_q^2/m for every n, where σ_q^2 is the per-query score variance ($\leq 1/4$ for scores in $[0,1]$). For any $\varepsilon < \sigma_q^2/m$ at the budget of Theorem 1(i), the estimator satisfies $\mathrm{MSE} \leq \varepsilon < \sigma_q^2/m$ for all sufficiently large n: query efficiency relative to the sample score in the sense of Definition 2, proving Corollary 1.

Contrast with paired DKPS. With $k_Q = \delta$, only identically-shared query pairs enter the double sum, so the U-statistic runs over the ρM_{ij} shared queries and the rate in Lemma 1 becomes $O_P\big((\rho M_{ij})^{-1/2}\big)$: DKPS's guarantee requires $\rho M_{ij} \to \infty$ and vacates in the unpaired limit $\rho \to 0$, where its distance is undefined (Section 3).

Proofs of Theorem 1(ii) and (iii). Split the reference models into a training set T and a validation set V with $|V| = n_v$. For each $\sigma \in \Sigma$, build the perspective space on T, embed any model outside T

out of sample by its Eq. 2 affinity row to T (the standard out-of-sample MDS coordinate), and let $\hat{y}_\sigma$ be nearest-neighbor regression on T's (perspective, score) pairs. The selected bandwidth is $\hat{\sigma} = \arg\min_{\sigma\in\Sigma} \hat{L}(\sigma)$ with $\hat{L}(\sigma) = \frac{1}{n_v}\sum_{i\in V}(\hat{y}_\sigma(f_i) - y_i)^2$. Conditional on T, the validation losses are i.i.d. across the draw of validation models and bounded in $[0,1]$, so by Hoeffding's inequality and a union bound over Σ, with probability at least $1-\delta'$, $\sup_{\sigma\in\Sigma}|\hat{L}(\sigma) - L(\sigma)| \le \tau = \sqrt{\log(2|\Sigma|/\delta')/2n_v}$, where $L(\sigma)$ is the MSE on a fresh target. On this event,

$$L(\hat{\sigma}) \;\le\; \hat{L}(\hat{\sigma}) + \tau \;\le\; \hat{L}(\delta) + \tau \;\le\; L(\delta) + 2\tau \;=\; \mathrm{MSE}(\mathrm{DKPS}) + 2\tau,$$

since the $k_Q = \delta$ member of the family coincides with paired DKPS on the shared queries with the same regression head (Section 3.1). Taking n_v large enough that $2\tau \le \varepsilon$ gives claim (iii) at every (m,ρ). If some $\sigma^\star$ has $\Delta = L(\delta) - L(\sigma^\star) > 0$, the same event gives $L(\hat{\sigma}) \le L(\sigma^\star) + 2\tau = \mathrm{MSE}(\mathrm{DKPS}) - \Delta + 2\tau$, strict for $2\tau < \Delta$. Finally, at $\rho = 0$ with nonatomic designs, $\Pr(q_j = q'_l) = 0$ for every pair of independently drawn queries, so almost surely every identity-matched sum is empty: any measurable completion of the DKPS member predicts as a function of the reference data alone, independent of the target, whence $\mathrm{MSE}(\mathrm{DKPS}) = v + \mathbb{E}[(\mathbb{E}y - \hat{y})^2] \ge v$, while under Assumptions 1–3 part (i) supplies a bandwidth attaining $\mathrm{MSE} \le \varepsilon$, so $\Delta \ge v - \varepsilon$ and the margin is at least $v - \varepsilon - 2\tau \ge v - 2\varepsilon$. In practice we use leave-one-model-out validation and a joint embedding (Section 3); the split protocol is analyzed for independence of the validation losses. $\square$